

Assignment-2

All questions and answers written below.

Some programs have been written in C, and some in C++.

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Question 1: What is a conditional statement in programming?

Answer:

A conditional statement is used to make decisions in a program based on a condition.

It executes different blocks of code depending on whether the condition is **true** or **false**.

Example:

```
if (age >= 18)
{
    cout << "Eligible to vote";
}
```

If the condition is true, the message will be displayed.

Question 2: What does an if statement do?

Answer:

An if statement checks a condition and executes a block of code only if the condition is **true**.

Example:

```
int age = 20;

if (age >= 18)
{
    cout << "Adult";
}
```

```
}
```

Since age is 20, the condition becomes true.

Output:

Adult

Question 3: What is the purpose of an else block?

Answer:

The else block is used to execute code when the condition in the if statement is **false**.

Example:

```
int age = 15;
```

```
if (age >= 18)
```

```
{
```

```
    cout << "Adult";
```

```
}
```

```
else
```

```
{
```

```
    cout << "Minor";
```

```
}
```

Output:

Minor

Question 4: When is an else block executed?

Answer:

An else block is executed when the condition inside the if statement becomes **false**.

Example:

```
int marks = 40;
```

```
if (marks > 50)
{
    cout << "Pass";
}
else
{
    cout << "Fail";
}
```

Since $40 > 50$ is false, the else block executes.

Output:

Fail

Question 5: What is a Boolean expression?

Answer:

A Boolean expression is an expression that gives only two results:

- true
- false

Example:

$10 > 5$

This expression returns **true** because 10 is greater than 5.

Another example:

$4 > 8$

This returns **false**.

Question 6: Write the syntax of a simple if statement.

Answer:

Syntax:

```
if (condition)
{
    // code to execute
}
```

Example:

```
int age = 18;

if (age >= 18)
{
    cout << "Eligible";
}
```

Question 7: What operator is commonly used to test equality?

Answer:

The == operator is used to test equality.

It checks whether two values are equal or not.

Example:

```
if (a == b)
{
    cout << "Equal";
}
```

Question 8: What does the modulus (%) operator return?

Answer:

The modulus (%) operator returns the **remainder** after division.

Example:

`10 % 3`

Result:

1

Because:

$10 \div 3 = 3 \text{ remainder } 1$

Question 9: How can you check if a number is even?**Answer:**

A number is even if it is completely divisible by 2.

Condition:

`number % 2 == 0`

Example:

```
int num = 8;
```

```
if (num % 2 == 0)
```

```
{
```

```
    cout << "Even Number";
```

```
}
```

Output:

Even Number

Question 10: How can you check if a number is odd?**Answer:**

A number is odd if it is **not divisible by 2**.

Condition:

`number % 2 != 0`

Example:

```
int num = 7;
```

```
if (num % 2 != 0)
```

```
{
```

```
    cout << "Odd Number";
```

```
}
```

Output:

Odd Number

Question 11: What is the purpose of else-if?**Answer:**

else-if is used to check multiple conditions one by one.

If the first condition is false, the program checks the next condition.

Example:

```
int marks = 75;
```

```
if (marks >= 90)
```

```
{
```

```
    cout << "Grade A";
```

```
}
```

```
else if (marks >= 70)
```

```
{
```

```
    cout << "Grade B";
```

```
}
```

```
else
```

```
{  
    cout << "Grade C";  
}
```

Output:

Grade B

Question 12: What is a nested if statement?

Answer:

A nested if statement means an if statement inside another if statement.

It is used to check multiple related conditions.

Example:

```
int age = 20;  
int marks = 80;  
  
if (age >= 18)  
{  
    if (marks >= 50)  
    {  
        cout << "Eligible";  
    }  
}
```

Output:

Eligible

Question 13: Can an if statement exist without an else?

Answer:

Yes, an if statement can exist without an else.

The code inside if executes only if the condition is true. If the condition is false, nothing happens.

Example:

```
int age = 20;
```

```
if (age >= 18)
{
    cout << "Adult";
}
```

Output:

Adult

Question 14: What is the output when a false condition is tested in an if statement without else?

Answer:

When the condition is false and there is no else block, **nothing is displayed**.

Example:

```
int age = 15;
```

```
if (age >= 18)
{
    cout << "Adult";
}
```

Since the condition is false, no output is shown.

Output:

No Output

Question 15: What is the role of braces {} in C/C++ conditionals?

Answer:

Braces {} are used to group multiple statements inside if, else, or loops.

They define the beginning and end of a block of code.

Example:

```
if (marks >= 50)
{
    cout << "Pass";
    cout << "Congratulations";
}
```

Without braces, only one statement executes.

Question 16: What does > mean?

Answer:

> means **greater than**.

It checks whether one value is greater than another.

Example:

10 > 5

Result:

true

Because 10 is greater than 5.

Question 17: What does < mean?

Answer:

< means **less than**.

It checks whether one value is smaller than another.

Example:

$5 < 10$

Result:

true

Because 5 is less than 10.

Question 18: What does $\geq$ mean?

Answer:

$\geq$ means **greater than or equal to**.

It checks whether a value is greater than or equal to another value.

Example:

$10 \geq 10$

Result:

true

Because both values are equal.

Question 19: What does $\leq$ mean?

Answer:

$\leq$ means **less than or equal to**.

It checks whether a value is smaller than or equal to another value.

Example:

$5 \leq 10$

Result:

true

Question 20: What does $\neq$ mean?

Answer:

$\neq$ means **not equal to**.

It checks whether two values are different.

Example:

`10 != 5`

Result:

true

Because 10 is not equal to 5.

Question 21: What is a switch statement?

Answer:

A switch statement is used to select one block of code from many choices.

It checks the value of a variable and executes the matching case.

Question 22: What is a case label in a switch statement?

Answer:

A **case label** is used to define a specific value inside a switch statement.

When the value matches, that block executes.

Example:

```
switch(day)
{
    case 1:
        cout << "Monday";
        break;

    case 2:
        cout << "Tuesday";
        break;
}
```

Here, case 1 and case 2 are case labels.

Question 23: Why is break used in switch?

Answer:

break is used to stop the execution of the switch statement after a matching case is found.

Without break, the program continues executing the next cases.

Example:

```
switch(day)
{
    case 1:
        cout << "Monday";
        break;

    case 2:
        cout << "Tuesday";
        break;
}
```

Question 24: What happens if break is omitted?

Answer:

If break is omitted, the program executes the next cases automatically. This is called **fall-through**.

Example:

```
int day = 1;
```

```
switch(day)
{
```

case 1:

```
cout << "Monday ";
```

case 2:

```
cout << "Tuesday ";
```

```
}
```

Output:

Monday Tuesday

Because there is no break, both cases execute.

Question 25: What is the default case in switch?

Answer:

The default case executes when none of the cases match.

It works like the else statement.

Example:

```
int day = 5;
```

```
switch(day)
```

```
{
```

```
case 1:
```

```
cout << "Monday";
```

```
break;
```

```
case 2:
```

```
cout << "Tuesday";
```

```
break;
```

default:

```
cout << "Invalid Day";
```

```
}
```

Output:

Invalid Day

Question 26: Can switch compare strings in C++?

Answer:

No, switch cannot compare strings directly in C++.

It usually works with:

- int
- char
- enum

Wrong Example:

```
string name = "Ali";
```

```
switch(name)
```

```
{
```

```
}
```

This will produce an error.

Question 27: Which statement is better for many constant choices: if-else or switch?

Answer:

switch is better when there are many constant choices because it makes the code cleaner and easier to understand.

Example:

For menu options:

```
switch(choice)
{
    case 1:
        cout << "Add";
        break;

    case 2:
        cout << "Delete";
        break;
}
```

Question 28: How do you test whether a value is positive?

Answer:

A number is positive if it is greater than 0.

Condition:

number > 0

Example:

```
int num = 5;
```

```
if(num > 0)
{
    cout << "Positive";
}
```

Question 29: How do you test whether a value is negative?

Answer:

A number is negative if it is less than 0.

Condition:

number < 0

Example:

```
int num = -5;
```

```
if(num < 0)
{
    cout<< "Negative";
}
```

Question 30: How do you test whether a value is zero?**Answer:**

A number is zero if its value is equal to 0.

Condition:

number == 0

Example:

```
int num = 0;
```

```
if(num == 0)
{
    cout<< "Zero";
}
```

Output:

Zero

Question 31: Write a condition to check if age is at least 18.

Answer:

"At least 18" means the age should be **greater than or equal to 18**.

Condition:

age >= 18

Example:

```
int age = 20;
```

```
if(age >= 18)
```

```
{
```

```
    cout << "Eligible to Vote";
```

```
}
```

Output:

Eligible to Vote

Question 32: Write a condition to check if marks are above 50.

Answer:

"Above 50" means marks should be **greater than 50**.

Condition:

marks > 50

Example:

```
int marks = 75;
```

```
if(marks > 50)
```

```
{
```

```
    cout << "Pass";
```

```
}
```

Output:

Pass

Question 33: What is decision making in programming?

Answer:

Decision making in programming means choosing different actions based on conditions.

Programs use conditional statements like:

- if
- if-else
- else-if
- switch

to make decisions.

Example:

```
int age = 17;
```

```
if(age >= 18)
{
    cout << "Adult";
}
else
{
    cout << "Minor";
}
```

Output:

Minor

Question 34: Can nested if statements have multiple levels?

Answer:

Yes, nested if statements can have multiple levels.

An if statement can be placed inside another if statement many times.

Question 35: What is logical AND (&&)?

Answer:

&& is called the **Logical AND operator**.

It returns **true only when both conditions are true**.

Question 36: What is logical OR (||)?

Answer:

|| is called the **Logical OR operator**.

It returns **true if at least one condition is true**.

Only one condition is true (marks >= 50), so output is shown.

Question 37: What is logical NOT (!)?

Answer:

! is called the **Logical NOT operator**.

It reverses the result of a condition:

- true becomes false
- false becomes true

Example:

```
int age = 20;
```

```
if(!(age < 18))
```

```
{
```

```
    cout << "Adult";
```

```
}
```

Output:

Adult

Question 38: How do you check if a year is divisible by 4?**Answer:**

A year is divisible by 4 if the remainder becomes 0.

Condition:

`year % 4 == 0`

Example:

```
int year = 2024;
```

```
if(year % 4 == 0)
```

```
{
```

```
    cout << "Divisible by 4";
```

```
}
```

Output:

Divisible by 4

Question 39: What is a leap year?**Answer:**

A leap year is a year that has **366 days instead of 365 days**.

It occurs every **4 years**.

Example:

2020, 2024, 2028

These are leap years.

Question 40: What is the first condition for a leap year?

Answer:

The first condition for a leap year is:

The year must be **divisible by 4**.

Condition:

```
year % 4 == 0
```

Question 41: How do you find the largest of two numbers?**Answer:**

The largest of two numbers can be found by comparing both numbers using an if-else statement.

C++ Program

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int num1, num2;
```

```
    cout << "Enter first number: ";
```

```
    cin >> num1;
```

```
    cout << "Enter second number: ";
```

```
    cin >> num2;
```

```
    if(num1 > num2)
```

```
    {
```

```
        cout << "Largest Number = " << num1;
```

```

    }

    else

    {

        cout << "Largest Number = " << num2;

    }


    return 0;

}

```

Output

```

PS E:\tranning\day2> g++ question41.cpp -o question41
>> ./question41
Enter first number: 20
Enter second number: 10
Largest Number = 20

```

Question 42: How do you find the largest of three numbers?

Answer:

The largest of three numbers can be found by comparing all numbers using if-else-if.

C++ Program

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int a, b, c;
```

```
    cout << "Enter three numbers: ";
```

```
    cin >> a >> b >> c;
```

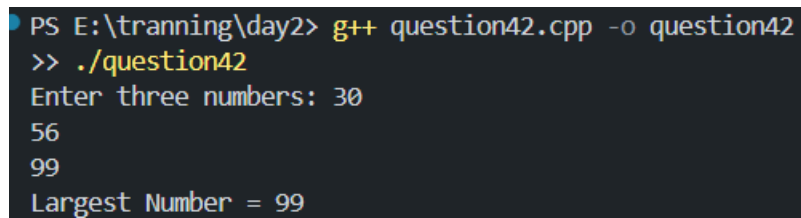
```

if(a >= b && a >= c)
{
    cout << "Largest Number = " << a;
}
else if(b >= a && b >= c)
{
    cout << "Largest Number = " << b;
}
else
{
    cout << "Largest Number = " << c;
}

return 0;
}

```

Output



```

PS E:\tranning\day2> g++ question42.cpp -o question42
>> ./question42
Enter three numbers: 30
56
99
Largest Number = 99

```

Question 43: What is an if-else-if ladder?

Answer:

An if-else-if ladder is used to check multiple conditions one by one. If one condition becomes true, the remaining conditions are skipped.

Question 44: What is the difference between if and switch?

Answer:

if	switch
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Used for conditions	Used for fixed choices
---------------------	------------------------

Can check ranges	Checks exact values
------------------	---------------------

More flexible	Cleaner for many choices
---------------	--------------------------

Question 45: Can switch use ranges directly?

Answer:

No, switch cannot use ranges directly. It only works with exact values. For ranges, if-else is used.

Question 46: What is fall-through in switch?

Answer:

Fall-through happens when break is not used in a switch statement, and the next case also executes automatically.

Question 47: What data type is usually used for conditions?

Answer:

The bool data type is usually used for conditions because it stores only true or false.

Question 48: Why are conditional statements important?

Answer:

Conditional statements are important because they help programs make decisions based on conditions.

Question 49: Give one real-life example of a conditional statement.

Answer:

ATM withdrawal system:

- If balance is enough → transaction successful

- Otherwise → insufficient balance

Question 50: Name four conditional structures covered.

Answer:

1. if statement
2. if-else statement
3. else-if ladder
4. switch statement